Summer School on Meta-Analysis

Meta-Analysis: Foundations and Recent Developments

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3–6 June 2026, University of Cagliari

Outline

1. **Before beginning**
2. **Intro**
3. **An illustrative example**
   - Aims and data
   - "First, take a look at data"
   - Fixed, random and mixed-effects meta-analysis
   - Evaluating heterogeneity of effect-sizes across studies
   - Explaining heterogeneity: Meta-Regression
   - Publication Bias
4. **Critical Issues**
5. **More**

From single studies to cumulative evidence

- *Ioannidis (2005). Why most published research findings are false*
- Scientific evidence is often noisy and fragmented
- A single study is rarely enough!
- Robust scientific conclusions require cumulative evidence

Increase meta-analytic reasoning among researchers

Fundamental steps in meta-analysis

1. **Identification**
   Find all of the pertinent studies addressing a clearly defined research question

2. **Selection**
   Select the right studies based on pre-specified eligibility criteria

3. **Abstraction**
   Abstract the relevant information from each study (sample size, effect sizes and other relevant study-level variables)

4. **Statistical analysis** ← **We are here!**
   Build the dataset. Summarize effect sizes. Evaluate and explain heterogeneity of effect-sizes across studies. Take into account issues related to "Publication Bias". Visualize and interpret results

5. **Publication of results**

Must-have references

Borenstein, Hedges, Higgins, & Rothstein. (2011)
Introduction to meta-analysis
John Wiley & Sons, Ltd.

Viechtbauer. (2010)
Conducting meta-analyses in R with the metafor package
Journal of Statistical Software, 36, 1-48

What is meta-analysis?

- "The analysis of analyses" *(Gene V. Glass, 1976)*

- Remember ...
  - "Meta-analytic summaries are all about weighted averages"
  - "Evaluating bias and heterogeneity are essential steps of meta-analysis"

(Stephanie Kovalchik, 2013)

More formally

1. A meta-analysis is the synthesis of:

   K compatible effects, Y_i

2. Giving greater weight to studies with:

   less variance, V_i (i.e., more precision: $W_i = \dfrac{1}{V_i}$)

Remember, V_i decreases as:

- *variability of data decreases*
- *sample size increases*

Type of effects

- An "effect" could be almost any aggregate statistic of interest (e.g., mean, proportion)
- In practice, the so-called measures of **effect-size** are often used:
    - Standardized mean difference (e.g., Cohen's d, Hedges'g)
    - Risk ratio, Odds ratio
    - Correlation
    - ...
- *N. B.* To conduct a meta-analysis you need to know/calculate the effect and its variance for each study

A real case-study

- During 1959 - 1988 there were **a total of 33 randomized trials** performed to assess the ability of streptokinase to prevent death following a heart attack.

- **Streptokinase**, a so-called clot buster which is administered intravenously, **was hypothesized to increase the likelihood of survival**

- **The trials all followed similar protocols**, with patients assigned at random to either treatment or placebo.
  The outcome, whether or not the patient died, **was the same in all the studies**

- In 1992 Lau et al. published a **meta-analysis that synthesized the results from the 33 studies**

Note: Relative Risk (RR) was used as a measure of effect size

Relative Risk as a measure of effect size

The *Relative Risk (RR)* or *Risk Ratio* is the ratio of the probability of an outcome in an exposed group to the probability of an outcome in an unexposed group.

In our case, for each study:

$$RR = \frac{\left(\frac{\text{Number of deaths among the treatment group}}{N_{\text{Treatment group}}}\right)}{\left(\frac{\text{Number of deaths among the placebo group}}{N_{\text{Placebo group}}}\right)}$$

- *RR close to 1* means that treatment does not affect the outcome
- $RR < 1$ means that the risk of the outcome is decreased by the treatment
- $RR > 1$ means that the risk of the outcome is increased by the treatment

Results (Forest plot adapted from Lau et al., 1992)

Study name	Statistics for each study				Sample size	Year
	Risk ratio	Lower limit	Upper limit	p-Value	Total	
Fletcher	0.229	0.030	1.750	0.155	23	1959
Dewar	0.571	0.196	1.665	0.305	42	1963
European 1	1.349	0.743	2.451	0.325	167	1969
European 2	0.703	0.534	0.925	0.012	730	1971
Heikinheimo	1.223	0.669	2.237	0.513	426	1971
Italian	1.011	0.551	1.853	0.973	321	1971
Australian 1	0.779	0.478	1.268	0.315	517	1973
Franfurt 2	0.457	0.252	0.828	0.010	206	1973
NHLBI SMIT	2.377	0.649	8.709	0.191	107	1974
Frank	0.964	0.332	2.801	0.946	108	1975
Valere	1.048	0.481	2.282	0.907	91	1975
Klein	2.571	0.339	19.481	0.361	23	1976
UK-Collab	0.922	0.609	1.394	0.699	595	1976
Austrian	0.608	0.417	0.886	0.010	728	1977
Australian 2	0.702	0.443	1.110	0.130	230	1977
Lasierra	0.282	0.034	2.340	0.241	24	1977
N Ger Collab	1.161	0.840	1.604	0.366	483	1977
Witchitz	0.813	0.263	2.506	0.718	58	1977
European 3	0.612	0.356	1.050	0.075	315	1979
ISAM	0.880	0.619	1.250	0.476	1741	1986
GISSI-1	0.827	0.749	0.914	0.000	11712	1986
Olson	0.429	0.041	4.439	0.477	52	1986
Baroffio	0.079	0.005	1.350	0.080	59	1986
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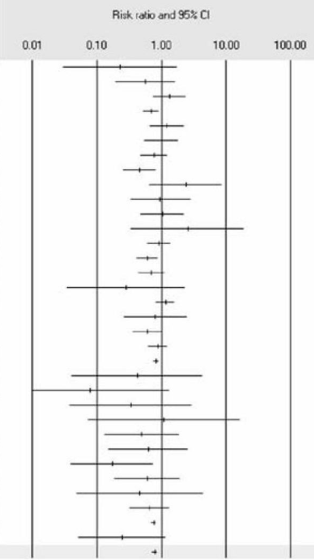
Wisenberg	0.244	0.051	1.164	0.077	66	1988
	0.794	0.724	0.870	0.000		



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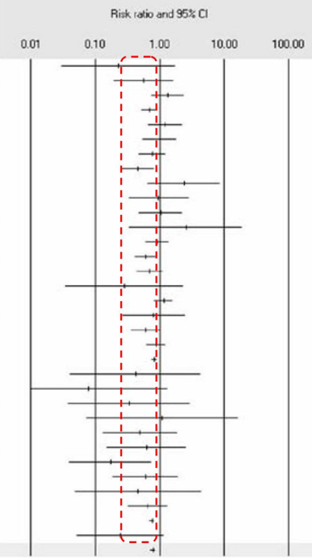

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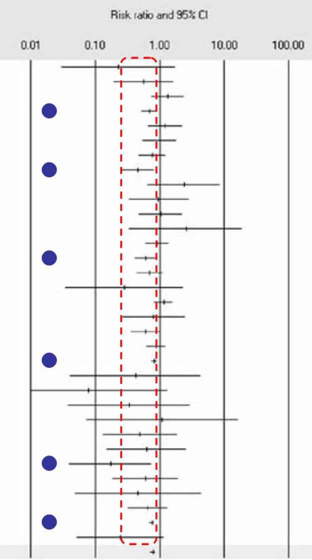

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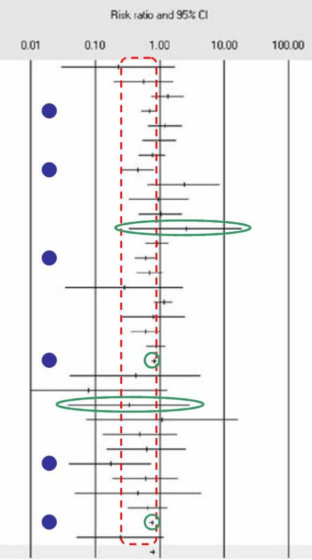

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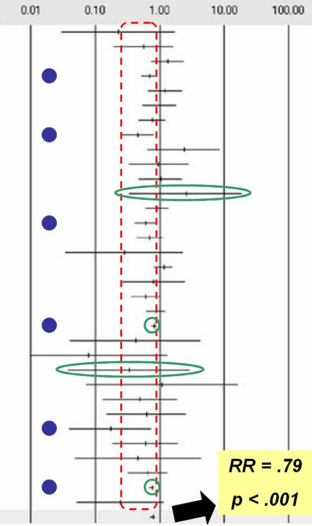
Risk ratio and 95% CI



RR = .79

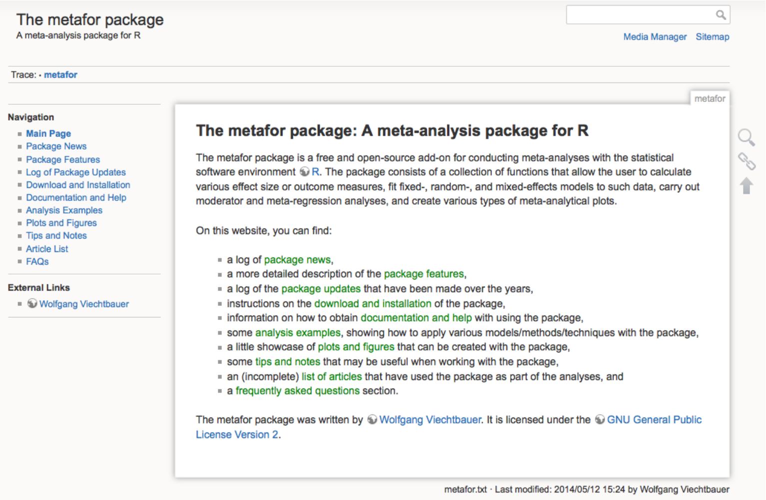
p < .001

“metafor”: An R package for meta-analyses

```
> install.packages("metafor") # install metafor (in R)
> library(metafor) # load metafor
> help(metafor) # ask for help (1)
> library(help= "metafor") # ask for help (2)
> citation("metafor") # how to cite
To cite the metafor package in publications, please use:

  Wolfgang Viechtbauer (2010).
  Conducting meta-analyses in R with the metafor package.
  Journal of Statistical Software, 36(3), 1-48. URL
  http://www.jstatsoft.org/v36/i03/.
```

The "metafor" website



http://www.metafor-project.org

… a very useful section: Analysis Examples

Login

Media Manager Sitemap

The metafor package

A meta-analysis package for R

Trace: • metafor • **analyses**

analyses

Navigation

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- Package Features
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- Documentation and Help
- **Analysis Examples**
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External Links

- Wolfgang Viechtbauer

Analysis Examples

The metafor package implements various meta-analytic models, methods, and techniques that have been described in the literature. The links below demonstrate how the models, methods, and techniques described in the respective articles/chapters can be applied via the metafor package. The articles are organized by topic (therefore, articles covering multiple topics may be listed multiple times). Alternatively, you can jump to the references at the bottom of the page for the list of articles in alphabetical order.

Fixed-, Random-, and Mixed-Effects Models

The articles below cover the standard fixed-, random-, and mixed-effects (meta-regression) models for meta-analysis.

- Berkey, C. S., Hoaglin, D. C., Mosteller, F., & Colditz, G. A. (1995). A random-effects regression model for meta-analysis. Statistics in Medicine, 14(4), 395-411.
- van Houwelingen, H. C., Arends, L. R., & Stijnen, T. (2002). Advanced methods in meta-analysis: Multivariate approach and meta-regression. Statistics in Medicine, 21(4), 589-624.
- Lipsey, M. W., & Wilson, D. B. (2001). Practical meta-analysis. Thousand Oaks, CA: Sage.

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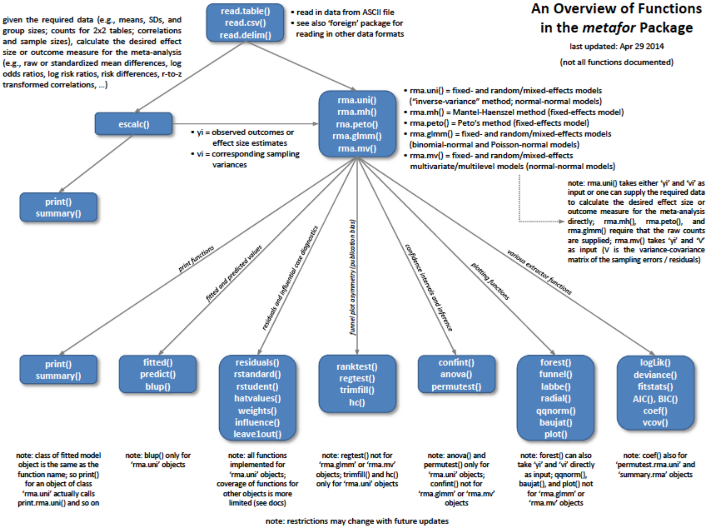
“metafor”: An overview



"metafor": The basic function rma()

```
> rma(yi, vi, method, ...)
```

where:

- `yi` = vector of effect-sizes
- `vi` = vector of associated variances
- `method` = model approach
- `...` = additional options

Aim

To evaluate the correlation between relational aggression and peer rejection

Card et al. (2008). Direct and Indirect Aggression During Childhood and Adolescence: A Meta-Analytic Review of Gender Differences, Intercorrelations, and Relations to Maladjustment. *Child Development*, 79, 1185 - 1229

Data (...adapted from Card, 2009)

	A	B	C	D	E	F	G
1	id	study	year	n	r	mean_age.years	published
2	1	Blachman	2003	228	0,525	9,20	no
3	2	Crick & Grotpeter	1995	491	0,198	9,40	yes
4	3	Crick et al.	1997	65	0,311	4,50	yes
5	4	Geiger	2003	458	0,554	8,00	no
6	5	Hawley et al.	2007	929	0,161	14,70	yes
7	6	Henington	1996	904	0,336	7,50	yes
8	7	Johnson	2003	74	0,396	6,00	no
9	8	Leff	1995	151	0,617	9,50	no
10	9	Miller	2001	150	0,557	16,00	no
11	10	Murray-Close & Crick	2006	590	0,575	9,00	yes
12	11	Nelson et al.	2005	180	0,039	4,80	yes
13	12	Ostrov	2008	139	0,358	3,60	no
14	13	Ostrov & Crick	2007	132	0,049	4,10	yes
15	14	Ostrov et al.	2004	60	0,000	4,60	yes
16	15	Pakaslahti & Keltikangas-Jarvinen	1998	839	0,326	14,50	yes
17	16	Phillipsen et al.	1999	262	-0,048	8,70	yes
18	17	Rys & Bear	1997	266	0,454	9,50	yes
19	18	Salmivalli et al.	2000	209	0,253	15,50	yes
20	19	Tomada & Schneider	1997	314	0,160	9,00	yes
21	20	Werner	2000	881	0,477	8,00	no
22	21	Werner & Crick	2004	517	0,469	8,00	yes
23	22	Zalecki & Hinshaw	2004	228	0,572	9,00	yes
24							

Available data sets: `datametaex1.xls` in Excel, `d.rda` in R

Data in R

```
> load("d.rda")
> head(d,4) # first 4 rows of data-set

  id               study year   n     r mean_age.years published
1  1            Blachman 2003 228 0.525            9.2        no
2  2 Crick & Grotpeter 1995 491 0.198            9.4       yes
3  3       Crick et al. 1997  65 0.311            4.5       yes
4  4              Geiger 2003 458 0.554            8.0        no

> str(d) # infos on data-set

'data.frame': 22 obs. of  7 variables:
 $ id            : num  1 2 3 4 5 6 7 8 9 10 ...
 $ study         : chr  "Blachman" "Crick & Grotpeter" "Crick et al." "Geiger" .
 $ year          : num  2003 1995 1997 2003 2007 ...
 $ n             : num  228 491 65 458 929 904 74 151 150 590 ...
 $ r             : num  0.525 0.198 0.311 0.554 0.161 0.336 0.396 0.617 0.557 0.
 $ mean_age.years: num  9.2 9.4 4.5 8 14.7 7.5 6 9.5 16 9 ...
 $ published     : Factor w/ 2 levels "no","yes": 1 2 2 1 2 2 1 1 1 2 ...
```

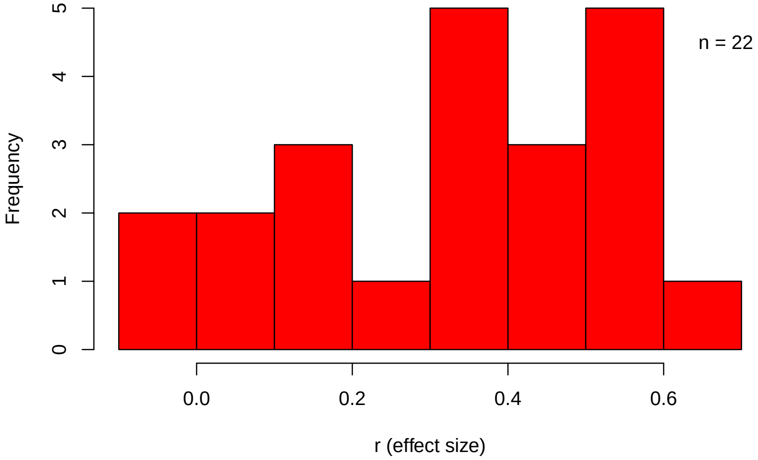
Descriptive statistics

```
> summary(d)

       id            study                 year            n        
 Min.   : 1.00   Length:22          Min.   :1995   Min.   : 60.0  
 1st Qu.: 6.25   Class :character   1st Qu.:1997   1st Qu.:150.2  
 Median :11.50   Mode  :character   Median :2002   Median :245.0  
 Mean   :11.50                      Mean   :2001   Mean   :366.7  
 3rd Qu.:16.75                      3rd Qu.:2004   3rd Qu.:510.5  
 Max.   :22.00                      Max.   :2008   Max.   :929.0  
       r              mean_age.years  published
 Min.   :-0.0480   Min.   : 3.600   no : 7    
 1st Qu.: 0.1703   1st Qu.: 6.375   yes:15    
 Median : 0.3470   Median : 8.850             
 Mean   : 0.3336   Mean   : 8.777             
 3rd Qu.: 0.5130   3rd Qu.: 9.475             
 Max.   : 0.6170   Max.   :16.000             
```

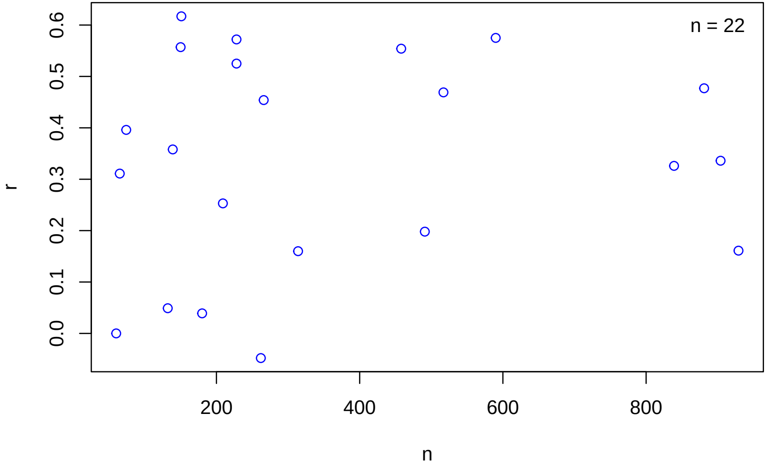
A first look at data (I)

```
> hist(d$r,col="red",main="",xlab="r (effect size)")
> text(.68,4.5,"n = 22")
> # COMMENTS?
```



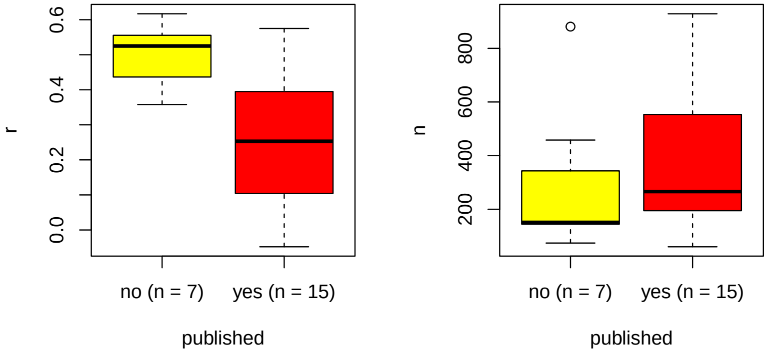
A first look at data (II)

```
> plot(d$n,d$r,col="blue",xlab="n",ylab="r")
> text(900,.6,"n = 22")
> # COMMENTS?
```



A first look at data (III)

```
> par(mfrow=c(1,2)) # ... (1)nrows x (2)ncols plots
> boxplot(d$r~d$published,col=c("yellow","red"),xlab="published",ylab="r",
+         names=c("no (n = 7)","yes (n = 15)"))
> boxplot(d$n~d$published,col=c("yellow","red"),xlab="published",ylab="n",
+         names=c("no (n = 7)","yes (n = 15)"))
> par(mfrow=c(1,1)) # (back to default)
> # COMMENTS?
```



Summarizing Effects: Basic Framework

1. **Normal Assumption**

$$Y_i \sim N(\theta, V_i)$$

2. **Summary effect size ($\hat{\theta}$) is a weighted average**

$$\hat{\theta} = \frac{\sum_i Y_i W_i}{\sum_i W_i} \quad , \quad W_i = \frac{1}{V_i}$$

$$Var(\hat{\theta}) = \frac{1}{\sum_i W_i}$$

where:

- Y_i and V_i = effect-size and associated variance of each study
- W_i = weight of each study

Modeling Approaches

- **Fixed-effects**
  - All studies share a *common true effect*
  - All observed variability reflects sampling error
    No between-study variability is taken into account
  - The summary effect is an *estimate of the common effect size*

- **Random-effects**
  - The true effects in the studies *are assumed to have been sampled* from a distribution of true effects.
  - *Two sources of variability* are taken into account:
    within-study variability and between-study variability.
  - The summary effect is an estimate of *the mean of all relevant true effects*

- **Mixed-effects (Meta-regression)**
  - *Study-level variables* are also taken into account to explain variability

Model formalizations: Fixed vs. Random

- **Fixed-effects model**

$$Y_i = \theta + \epsilon_i$$

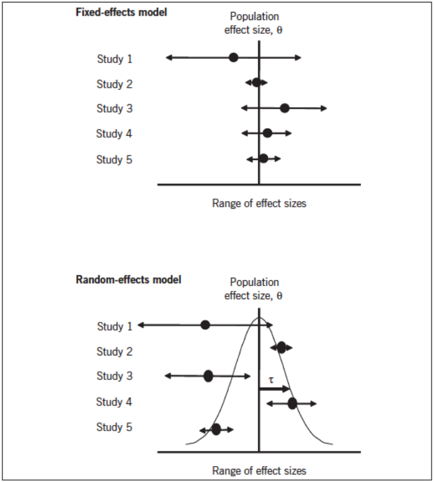
$$\epsilon_i \sim N(0, V_i)$$

- **Random-effects model**

$$Y_i = \theta + \theta_i + \epsilon_i$$

$$\theta_i \sim N(0, \tau^2) \quad , \quad \epsilon_i \sim N(0, V_i)$$

Conceptual representation: Fixed vs. Random



Adapted from Card, 2009

Fixed vs. Random: Some remarks (1)

- The *fixed-effects model* is a description of the K studies
- The *random-effects model* regards the K studies as a sample of a larger universe of studies
- In general, for observational studies, the *random-effects model* should be preferred

Fixed vs. Random: Some remarks (2)

- A fixed-effect meta-analysis estimates a single effect that is assumed to be common to every study, while a random-effects meta-analysis estimates the mean of a distribution of effects
- Study weights are more balanced under the random-effects model than under the fixed-effect model. Large studies are assigned less relative weight and small studies are assigned more relative weight as compared with the fixed-effects model
- The standard error of the summary effect and (it follows) the confidence intervals for the summary effect are wider under the random-effects model than under the fixed-effect model

Mixed-effects model formalization

$$Y_i = \boldsymbol{X}\boldsymbol{\beta} + \theta + \theta_i + \epsilon_i$$

$$\theta_i \sim N(0, \tau^2) \quad , \quad \epsilon_i \sim N(0, V_i)$$

where:

$\boldsymbol{X}$ = design matrix of study-level predictor variables

Before running a meta-analysis: Transformation of effect-sizes

- As meta-analysis is based on normality, in some cases it is preferable to normalize effect-sizes via appropriate transformations
- For example, the sampling distribution of correlation coefficients (r) is not normally distributed. In this case, before running a meta-analysis, the correlation coefficients are usually transformed using the *Fisher's z transformation*
- Typical practice is to compute r for each study, convert these to z_r for meta-analytic analysis, and then convert results back to r for reporting

Fisher's z transformation

$$z_{r_i} = .5 \times \ln\left(\frac{1 + r_i}{1 - r_i}\right)$$

$$Standard\,Error(z_{r_i}) = \frac{1}{\sqrt{n_i - 3}} \quad , \quad n_i = \text{sample size of study } i$$

$$Variance(z_{r_i}) = \left(Standard\,Error(z_{r_i})\right)^2$$

...and now, we are ready for our data!

Fisher's transformation applied to data

```
> library(psych) # Various psychometric utilities, see ?psych
> d$zi = fisherz(d$r) # Fisher transformed values
> d$z.vi = 1/(d$n-3)  # Associated variances
> #
> # Values of first 5 studies
> round( (cbind(id=d$id,r=d$r,zi=d$zi,z.vi=d$z.vi)[1:5,]), 3)

     id     r    zi  z.vi
[1,]  1 0.525 0.583 0.004
[2,]  2 0.198 0.201 0.002
[3,]  3 0.311 0.322 0.016
[4,]  4 0.554 0.624 0.002
[5,]  5 0.161 0.162 0.001
```

Fixed- and Random-effects meta-analysis

```
> library(metafor) # load "metafor"
> # Fixed-effects meta-analysis
> m.fixed <- rma(yi = d$zi, vi = d$z.vi, method = "FE")
> #
> # Random-effects meta-analysis
> #  with Restricted Maximum Likelihood estimation
> m.random <- rma(yi = d$zi, vi = d$z.vi, method = "REML")
> #
> # See ?rma, for options, returned results and details
```

Results

```
> # Fixed meta-analysis
> # summary effect size
> round(m.fixed$b , 3)

        [,1]
intrcpt 0.38

> # 95% confidence interval
> round(c(m.fixed$ci.lb , m.fixed$ci.ub) , 3)

[1] 0.358 0.402

> # Random meta-analysis
> # summary effect size
> round( m.random$b , 3)

         [,1]
intrcpt 0.368

> # 95% confidence interval
> round(c(m.random$ci.lb , m.random$ci.ub) , 3)

[1] 0.270 0.466

> # COMMENTS?
```

Back-transformed results

```
> # Results in terms of correlation coefficient
> #
> # Fixed meta-analysis
> round( fisherz2r(m.fixed$b) , 3)

         [,1]
intrcpt 0.363

> round( fisherz2r(c(m.fixed$ci.lb , m.fixed$ci.ub)) ,3)

[1] 0.344 0.382

> #
> # Random meta-analysis
> round( fisherz2r(m.random$b) , 3)

         [,1]
intrcpt 0.352

> round( fisherz2r(c(m.random$ci.lb , m.random$ci.ub)) , 3)

[1] 0.263 0.435

> #
> # COMMENTS?
```

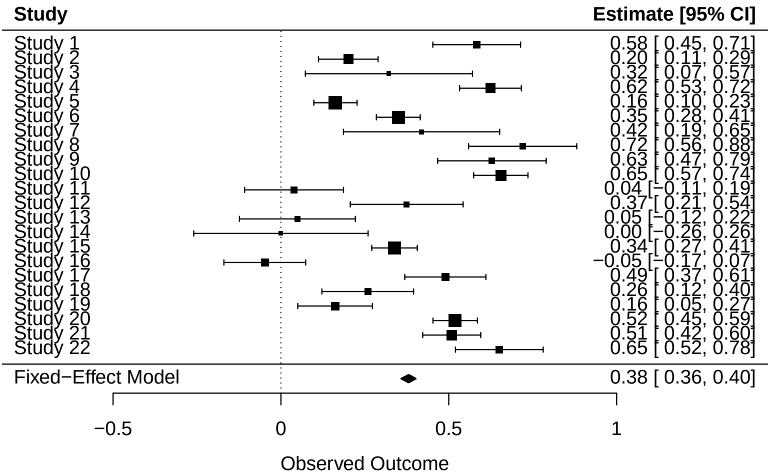
Visualizing results: The forest plot

The forest plot:

- Is a plot of effect sizes and their precisions. Usually:
  - The observed effects are drawn as boxes proportional to the precision of the estimates. The associated confidence intervals are also presented
  - The summary effect-size is presented as a diamond, with the outer edges indicating the confidence intervals
- Is the most common way to report the results of a meta-analysis
- Can help identify patterns across effects
- Can help spot large variation in effects or possible outliers

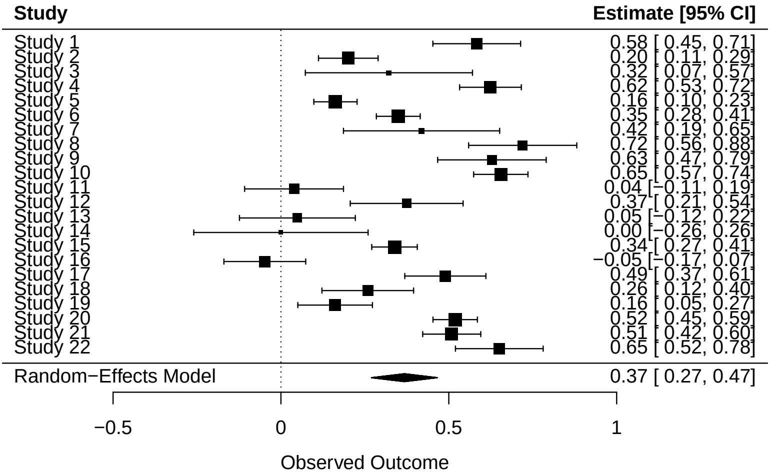
Forest plot: Fixed-effects meta-analysis

```
> forest(m.fixed)
> # (Default plot ... see ?forest.rma for AWESOME customizations!)
> # COMMENTS?
```



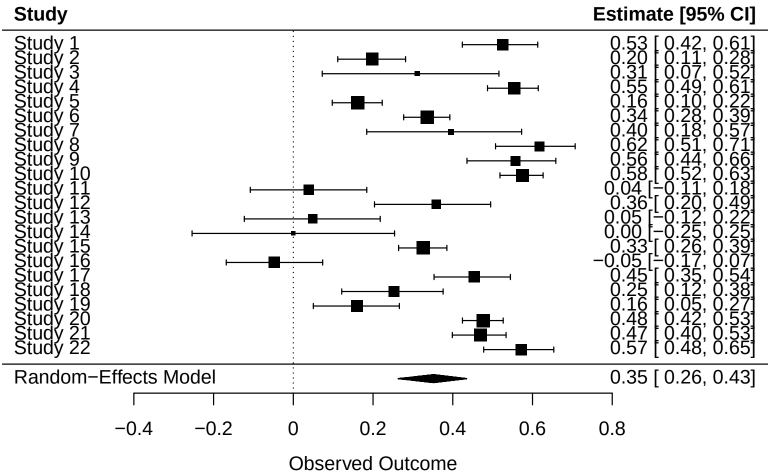
Forest plot: Random-effects meta-analysis

```
> forest(m.random)
> # COMMENTS?
```



Forest plot: Back-transformed values

```
> forest(m.random, transf = fisherz2r)
> # forest plot in terms of correlation coefficients
> # COMMENTS?
```



Evaluating heterogeneity of effect-sizes across studies

- Forest Plot
- Q test for heterogeneity
- Indices Of Heterogeneity: τ^2, τ, Higgins' I^2

Q test for heterogeneity

- Q is the weighted sum of squared differences between individual study effects and the summary effect.
  Weights are those used in the meta-analysis
- Larger values of Q reflect greater between-study heterogeneity
- $Q \sim \chi^2{}_{(K-1)}$ with K = number of studies
  If the associated $p - value$ is significant,
  "you can reject the null hypothesis of homogeneity ..."

Q test for heterogeneity: Warnings

- If there are few studies in the meta-analysis, the $Q - test$ is likely **underpowered** for detecting true heterogeneity
- Conversely, $Q - test$ has **too much power** if the number of studies is large
- **N.B**. $Q - test$ should be evaluated together with forest plot and indices of heterogeneity

Heterogeneity indices

- $\boldsymbol{\tau^2}$: estimate of between-study variance
- $\boldsymbol{\tau}$: estimate of between-study standard deviation
- **Higgins'** $\boldsymbol{I^2}$: estimates, in percentage, how much of the total variability in the effect size estimates can be attributed to heterogeneity among the true effects
  $(\tau^2 = 0 \longrightarrow I^2 = 0)$

Note: I^2 should not be interpreted mechanically. Recent discussions emphasize that *prediction intervals* are often more informative for understanding the dispersion of true effects across studies (see Borenstein, 2023; Supplemental Materials). Prediction intervals will also be discussed during the practical sessions.

It's your turn

```
> summary(m.random)

Random-Effects Model (k = 22; tau^2 estimator: REML)

  logLik  deviance       AIC       BIC      AICc
  0.5830   -1.1659    2.8341    4.9231    3.5007

tau^2 (estimated amount of total heterogeneity): 0.0498 (SE = 0.0169)
tau (square root of estimated tau^2 value):      0.2232
I^2 (total heterogeneity / total variability):   94.62%
H^2 (total variability / sampling variability):  18.59

Test for Heterogeneity:
Q(df = 21) = 322.5654, p-val < .0001

Model Results:

estimate      se    zval    pval   ci.lb   ci.ub
  0.3677  0.0499  7.3607  <.0001  0.2698  0.4656  ***

---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Explaining heterogeneity: Meta-Regression

- To explain heterogeneity, study-level predictor variables can be specified
- In this case, a mixed-effects model (i.e., Meta-Regression) will be performed
- In `rma()`, study-level predictors (categorical and/or quantitative) can be specified via the `mod` argument

Let's consider the effect of "mean age of participants" and "publication" on our data . . . $\Longrightarrow$

Mixed-effects meta-analysis

```
> m.mixed<-rma (yi = zi, vi = z.vi,
+           mods = cbind(mean_age.years,published),
+           method = "REML", data = d)
```

Results of mixed-effects meta-analysis

```
> summary(m.mixed)

Mixed-Effects Model (k = 22; tau^2 estimator: REML)

  logLik  deviance       AIC       BIC      AICc
  3.7568   -7.5137    0.4863    4.2641    3.3435

tau^2 (estimated amount of residual heterogeneity):     0.0351 (SE = 0.0129)
tau (square root of estimated tau^2 value):             0.1874
I^2 (residual heterogeneity / unaccounted variability): 92.11%
H^2 (unaccounted variability / sampling variability):   12.68
R^2 (amount of heterogeneity accounted for):            29.49%

Test for Residual Heterogeneity:
QE(df = 19) = 233.7530, p-val < .0001

Test of Moderators (coefficients 2:3):
QM(df = 2) = 10.1900, p-val = 0.0061

Model Results:

                estimate      se     zval    pval    ci.lb    ci.ub
intrcpt           0.7094  0.1900   3.7333  0.0002   0.3370   1.0818  ***
mean_age.years    0.0146  0.0122   1.1932  0.2328  -0.0094   0.0385
published        -0.2789  0.0923  -3.0227  0.0025  -0.4598  -0.0981   **

---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Publication Bias in meta-analysis

- **Publication bias**: studies included in the meta-analysis differ systematically from all studies that should have been included

- Typically, studies with larger than average effects are more likely to be published, and this can lead to an **upward bias** in the summary effect

(Borenstein, Hedges, Higgins, & Rothstein, 2011)

STATISTICAL management of Publication Bias (in brief)

- **Funnel Plot**
- **Trim and fill method**

Funnel Plot

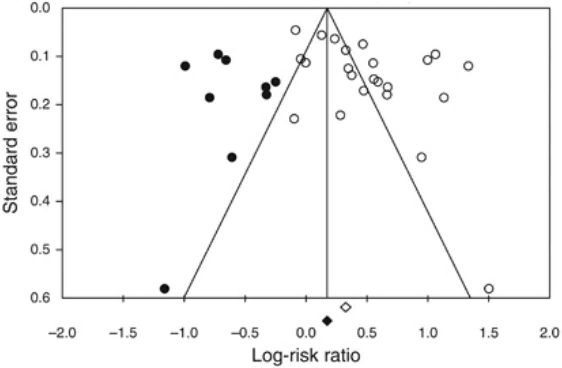
- It's a kind of scatterplot of the effects against a measure of study precision.
- In the absence of publication bias, it assumes that the largest studies will be plotted near the average, and smaller studies will be spread evenly on both sides of the average, creating a funnel-shaped distribution
- Asymmetry (gaps) in the funnel may be indicative of publication bias

Trim and fill method

The trim-and-fill method:

- is a nonparametric data augmentation technique
- can be used to estimate the number of studies missing from a meta-analysis due to the suppression of the most extreme results on one side of the funnel plot
- augments the observed data so that the funnel plot is more symmetric
- should not be regarded as a way of yielding a more "valid" estimate of the summary effect, but as a way of examining the sensitivity of the results to possible forms of publication bias

Example: Funnel Plot with Trim and Fill Method

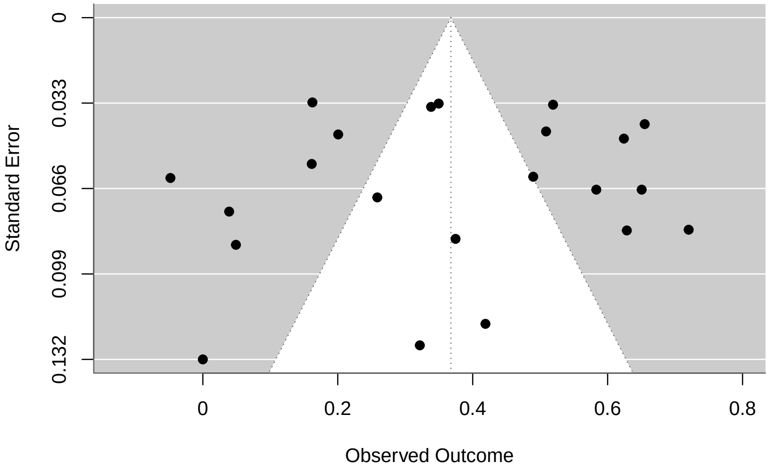


- In white, observed effects and associated summary effect
- In black, hypothetical studies using the trim-and-fill method and the updated summary effect

Janarthanan et al. (2012). *Clostridium difficile-associated diarrhea and proton pump inhibitor therapy: a meta-analysis.* The American Journal of Gastroenterology, 107, 1001-1010. Impact Factor = 7.28

Back to our data: Funnel plot

```
> funnel(m.random)
> # COMMENTS?
```



Back to our data: Trim and Fill Method

```
> trimfill(m.random)

Estimated number of missing studies on the right side: 0 (SE = 2.8869)

Random-Effects Model (k = 22; tau^2 estimator: REML)

tau^2 (estimated amount of total heterogeneity): 0.0498 (SE = 0.0169)
tau (square root of estimated tau^2 value):      0.2232
I^2 (total heterogeneity / total variability):   94.62%
H^2 (total variability / sampling variability):  18.59

Test for Heterogeneity:
Q(df = 21) = 322.5654, p-val < .0001

Model Results:

estimate      se     zval    pval   ci.lb   ci.ub
  0.3677  0.0499   7.3607  <.0001  0.2698  0.4656  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

> # Note: to create a trim-and-fill funnel plot:
> #  funnel(trimfill(m.random))
```

Other innovative approaches (hints)

- Carter, E. C., Schonbrodt, F. D., Gervais, W. M., & Hilgard, J. (2019). Correcting for bias in psychology: A comparison of meta-analytic methods. *Advances in Methods and Practices in Psychological Science*, 2(2), 115-144.

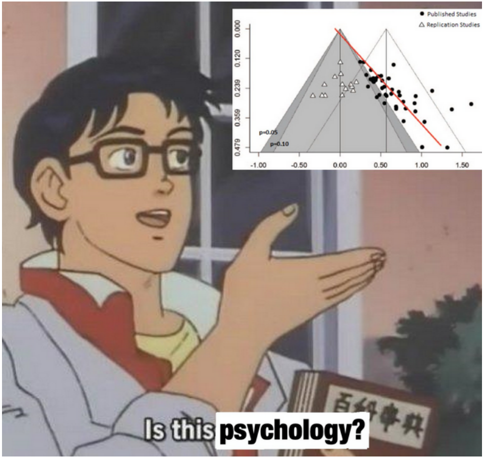
- Haaf, J. M. (2020). Conventional Publication Bias Correction Methods.
  Pre-print: https://psyarxiv.com/gv4tw/

. . . a recent tweet on meta-analysis

John Protzko
@JProtzko

Does cognitive training transfer to cog abilities in the elderly? YES according to a 2019 meta-analysis (europepmc.org/abstract/med/3...)...or marginally/maybe according to a 2019 meta-analysis (sciencedirect.com/science/articl...)...or NO according to a 2019 meta-analysis (sciencedirect.com/science/articl...).

. . . and another one

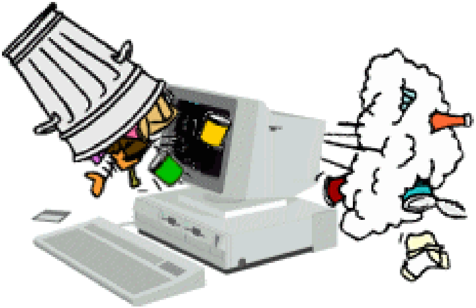


NO, THIS IS BAD PSYCHOLOGY!!!

Always remember GIGO

GIGO!

Garbage In $\Longrightarrow$ Garbage Out!



An "ideal way", you should focus on !, to plan a meta-analysis

"The key result of a review on publication bias is that meta-analysis may be overly trusted.

While it is important to aggregate results from smaller studies to obtain more robust results, a meta-analysis of the literature may not be the ideal way.

Alternatives are **large-scale many-labs or replication projects**, *where transparency and openness guidelines naturally limit publication bias, QRPs (questionable research practices) [& QMPs (questionable measurement practices)]."*

(Julia Haaf, 2020)

Take-home message

(. . . like all statistical models [see, George Box ~ 1976] . . .)

All meta-analyses are wrong,
but some are useful!

If you do not simulate it, you have not understood it ;)

Advances in Methods and Practices in Psychological Science

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Open access | Research article | First published online February 13, 2024

Understanding Meta-Analysis Through Data Simulation With Applications to Power Analysis

Filippo Gambarota and Gianmarco Altoè View all authors and affiliations

All Articles | https://doi.org/10.1177/25152459231209330

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Abstract

Meta-analysis is a powerful tool to combine evidence from existing literature. Despite several introductory and advanced materials about organizing, conducting, and reporting a meta-analysis, to our knowledge, there are no introductive materials about simulating the most common meta-analysis models. Data simulation is essential for developing and validating new statistical models and procedures. Furthermore, data simulation is a powerful educational tool for understanding a statistical method. In this tutorial, we show how to simulate equal-effects, random-effects, and metaregression models and illustrate how to estimate statistical power. Simulations for multilevel and multivariate models are available in the Supplemental Material available online. All materials associated with this article can be accessed on OSF (https://osf.io/54djn/).

If you do not simulate it, you have not understood it.

Available here:
https://journals.sagepub.com/doi/full/10.1177/25152459231209330

... some useful papers with Further Related Topics, Data, Codes and Supplemental Materials

- Dumitru, E. P., Cardoș, R. A., Milea, I., Gambarota, F., Altoè, G., David, D. O., & Cristea, I. A. (2025). A meta-analysis of the association of death anxiety with psychological distress and psychopathology. Nature Human Behaviour, 1–11.
  keywords: collaboration is the key :)
- Carnevali, L., Valori, I., Mason, G., Altoè, G., & Farroni, T. (2024). Interpersonal motor synchrony in autism: a systematic review and meta-analysis. Frontiers in Psychiatry.
  keywords: prediction intervals
- Montuori, C., Gambarota, F., Altoè, G., & Arfé, B. (2023). The cognitive effects of computational thinking: A systematic review and meta-analytic study. Computers & Education.
  keywords: randomized controlled trials, exploratory multiverse meta-analysis
- Benavides-Varela, S., Callegher, C. Z., Fagiolini, B., Leo, I., Altoè, G., & Lucangeli, D. (2020). Effectiveness of digital-based interventions for children with mathematical learning difficulties: A meta-analysis. Computers & Education.
  keywords: randomized controlled trials, sensitivity analysis
- Talamini, F., Altoè, G., Carretti, B., & Grassi, M. (2017). Musicians have better memory than nonmusicians: A meta-analysis. PLOS ONE.
  one possible trajectory: meta-analysis → multi-lab replication :)

  Grassi, M., Talamini, F., Altoè, G., et al., many others! (2025). Do musicians have better short-term memory than nonmusicians? A multilab study. Advances in Methods and Practices in Psychological Science.

. . . last suggestion

USE graphical representations!

"There is no statistical tool that is as powerful as a well-chosen graph." (Chambers et al., 1983)

Questions

any questions!?

gianmarco.altoe@unipd.it

PSICOSTAT

https://psicostat.dpss.psy.unipd.it/